

The impact of Generative AI on changing organizational workflow structures on AstraZeneca employees

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Abstract

This research focuses on Generative Artificial Intelligence's (GenAI) impact through personal and collaborative workflows in AstraZeneca. GenAI has its benefits and downsides; this research explores the effects of time efficiency, mental challenges, problem-solving, collaboration, and technical changes. The research uses qualitative interpretative research with semi-structured interviews that were conducted online and in person with three AstraZeneca members. The findings reveal that GenAI has technical and research workflows that help speed up processes, shifting employee roles from manual production to high-speed curation but still requiring human judgment and domain expertise.

Keywords

Generative Artificial Intelligence, Platform Strategy, Organizational Structure, Workflows, and Collaborative Processes.

Introduction

As of 2026, GenAI has been increasingly integrated into workplace applications, as enterprises try to maximize employees' working efficiency and create a more optimal working process. GenAI can create new content such as text, images, code, and summaries based on existing data (NIST, 2024). Research also shows that GenAI can improve workplace productivity and help employees complete tasks faster, especially in knowledge-based work. As adoption is growing quickly, many businesses apply GenAI with the mindset of "*follow or be left behind*" (Bick, Blandin & Deming, 2024). However, the use of GenAI in organizational workflows differs across industries, and there is no single standard way to implement it successfully. The effectiveness of GenAI depends on the business context, task type, employee skills, and human-AI collaboration (WEF & PwC, 2024).

This paper discusses the research question: "**How does Generative AI impact organizational workflow structures and employees at AstraZeneca?**" This research focuses on AstraZeneca, a technology-driven pharmaceutical enterprise in Sweden, where GenAI is used for both general office tasks and in complex workflows like drug discovery, data analysis, statistical programming, and clinical documentation. Exploring how GenAI is used in daily tasks makes the case interesting because pharmaceutical workflows require both efficiency and strong human validation (McKinsey, 2024). The research gap lies in the fact that existing studies primarily examine the productivity effects of GenAI across general workplace environments (Brynjolfsson et al., 2023). While these studies demonstrate improvements in productivity and work efficiency, there is still limited understanding of how GenAI is integrated into everyday workflows within highly regulated pharmaceutical organizations and how these changes affect employees' roles and work practices.

Previous research and theoretical framework

Previous research

GenAI is increasingly used in organizations to support workflows, including coding, documentation, data analysis, and information processing. Some previous research shows that GenAI can improve productivity

and help employees complete tasks faster, especially in repetitive tasks (Brynjolfsson, Li & Raymond, 2023). However, rapid adoption does not automatically lead to organizational transformation. GenAI must be integrated carefully into existing workflows, supported by employee skills, governance, and strategic implementation (Bick, Blandin & Deming, 2024; Dillon et al., 2025; WEF & PwC, 2024).

In pharmaceutical companies such as AstraZeneca, GenAI can support specialized workflows such as drug discovery, data analysis, statistical programming, and clinical documentation. However, as these workflows are scientific and highly regulated, GenAI still requires human judgement and domain expertise. Therefore, this research applies Job Crafting Theory to understand how employees adapt their tasks, work relationships, and understanding of work when using GenAI. This framework helps analyse how GenAI is integrated into AstraZeneca's workflows to discover the research question "How does Generative AI impact organizational workflow structures and employees at AstraZeneca?", while also showing where human expertise remains necessary.

Theoretical framework

This research uses Job Crafting Theory as the main theoretical framework. It explains that employees are active crafters of their work, as they can change the task, relational, and cognitive boundaries of their jobs (Wrzesniewski & Dutton, 2001). This framework fits the research as it examines how employees at AstraZeneca adapt their workflows when using GenAI.

The analysis uses three dimensions: (i) **Task Crafting** refers to how employees change the way tasks are done. In this research, it is used to analyse how employees use GenAI for automation, coding support, documentation, data analysis, molecule generation, and other workflow-related tasks. (ii) **Relational Crafting** refers to how employees change the way they interact with others. In this research, it is used to analyse whether GenAI changes collaboration between teams. (iii) **Cognitive Crafting** refers to how employees change the way they view their work. In this research, it is used to analyse how employees understand GenAI as a support tool, how they judge AI reliability, and how they decide which tasks should remain human-controlled.

Using Job Crafting Theory allows this research to move beyond asking whether GenAI improves productivity. Instead, it helps explain how employees actively adapt and reshape their workflows around GenAI. This is important because the value of GenAI in AstraZeneca depends not only on the technology itself, but also on how employees use and integrate it into their daily work.

Methodology

This research discusses a qualitative interpretative research approach by gathering data through semi-structured interviews with three employees at AstraZeneca to understand employees' own experiences of using GenAI in daily work. One of the interviewees is doing a graduate program and working full-time for AstraZeneca, one of them is doing their master's thesis and working part-time, and one is working at the R&D department as an IT Business Partner. They are all employed by AstraZeneca and consented to the interview data being used.

In this study, organizational workflows refer to the sequence of tasks, interactions, and decision-making processes through which employees complete their work. These workflows include activities such as coding, documentation, data analysis, project management, laboratory-related processes, and decision support. The interview questions were designed to address the research question and were structured around the three dimensions of Job Crafting Theory (Wrzesniewski & Dutton, 2001).

Data collection

This research focuses on AstraZeneca, specifically examining its workflow operations in Gothenburg, Sweden. The research was conducted through one in-person interview and three online interviews. Each interviewee has different roles, one in an internship master thesis in the Molecular AI department,

another in a graduate program for research and development for testing methods for drugs, and the third one in acting as bridge between the IT team and R&D team. The interviews were between 25 and 35 minutes, and it was semi-structured, which allowed for flexibility. The interviewees were selected through availability and connections, not specifically for their role, thus it was a convenience sampling. The participants have different positions, which can lead to different experiences with the use of GenAI in their field. Interviewing participants across varying roles is problematic because the lack of a shared baseline workflow prevents meaningful comparison and forces a superficial analysis.

Data analysis

The interviews were recorded and then transcribed; then they were labelled based on in which employees actively modify and reshape their tasks, relationships, and work processes to work more effectively (Wrzesniewski & Dutton, 2001). Afterwards, the interviews were then categorized through different sub-themes which then represent different parts of answers for the research question. The main sub-themes are: **Workflow Changes, Organizational Friction, Cultural Friction, and Future Workflows**, which serve as indicators of the three overarching Job Crafting dimensions within the workspace.

Lastly, the analysis was done through coding with the help of Google spreadsheets and ChatGPT by labelling the evidence from the transcript with three Job Crafting dimensions and four sub-themes by linking it with the initial code. The prompt for coding was “Code the interview transcript using Job Crafting Theory. For each relevant quote, identify the initial code, Job Crafting dimension, sub-theme, and meaning. Use the three dimensions: Task Crafting, Relational Crafting, and Cognitive Crafting, and the four sub-themes: Workflow Changes, Organizational Friction, Cultural Friction, and Future Workflows”. This assisted us in organising the transcription data by finding keywords through the coding and labelling them with the dimensions and sub-themes.

After that, the AI-assisted coding results were manually checked by the researchers. The evidence, initial codes, Job Crafting dimensions, and sub-themes were reviewed to ensure they matched the transcript meaning. Each quote was checked to see whether they matched the meaning of the interviewees' answer. Any unclear labels were adjusted manually.

Result

The coding results show that **Task Crafting** and **Cognitive Crafting** are the two dominant dimensions in all interviews' transcription, which accounts for the majority of coded evidence (78% in Interview 1, 91% in Interview 2, and 72% in Interview 3). It implies that GenAI is mainly affecting AstraZeneca workflows in two ways: (1) employees change how tasks are done through AI-supported coding, automation, data analysis, molecule generation, and documentation, and (2) employees change how they think about work, especially AI reliability, responsible use, data privacy, hallucinations, and human validation. In the meantime, **Relational Crafting** appears less frequently, which suggests that GenAI has not yet strongly transformed collaboration across teams.

Task Crafting

Interviewees consistently described how GenAI transformed daily work activities through automation, coding support, information retrieval, and data analysis. The Interviewee 1 stated that AI had become central to their work: “For me, 100% coding” and that “Code that would take me months to do it in 5 minutes”. Similarly, the Interviewee 3 explained that developers were increasingly using “prompt coding” and described a debugging task that was reduced from “maybe a week” to “one day” through AI assistance. These findings suggest that GenAI is accelerating workflow execution by reducing time spent on repetitive and technical tasks.

Cognitive Crafting

Interviewees emphasized that AI outputs still require human judgement, validation, and responsible use. The Interviewee 1 stated, “For me, at the end it’s my job,” while also noting that “Whatever I produce then has to have a chemist saying ‘this makes sense’”. Concerns about hallucinations and reliability were also common. Interviewees referred to “hallucinations and human in the loop” training (Interviewee 3) and highlighted the need for “a higher level of fact checking” (Interviewee 2). These responses indicate that employees are adapting their understanding of work by treating AI as a support tool rather than an autonomous decision-maker.

Relational Crafting

Relational Crafting appeared less frequently in all interviews. However, participants described collaboration between AI specialists, scientists, and business teams. The Interviewee 1 explained: “We propose new molecules, they try those molecules in the lab. They give the results back to us, we improve our models”. The Interviewee 3 highlighted collaboration between platform teams, engineers, and scientists during AI-related projects. At the same time, several barriers were identified. Interviewees reported that “there’s not enough communication between departments inside AstraZeneca” (Interviewee 1) and pointed to challenges in AI adoption, training, and knowledge sharing across teams (Interviewee 3). This suggests that while GenAI supports cross-functional collaboration, its impact on organizational relationships remains less significant than its impact on tasks and employee perceptions.

Overall, GenAI is not simply replacing human work. Instead, it is being used as a workflow support tool that helps employees speed up technical tasks while still requiring human judgment and domain expertise.

Discussion

The findings show that the use of generative AI is a new modern-day tool that comes with positives that employees adapt organizational workflows through selective use of GenAI, especially in technical and repetitive tasks, but with some negative effects.

Effects of GenAI

A primary theme that emerged from the data was the presence of technical challenges. These challenges can be found in training reliable chemistry models. From the findings, we discovered that public data is inherently “noisy” due to the various unmonitored laboratory conditions. In the Job Crafting Theory, the first dimension is “Task Crafting”. Employees have a quicker and more convenient way to finish their work through the use of GenAI.

Secondly, there is another side effect observed in the research workflow regarding documentation. The findings suggest that identifying and mitigating hallucinations is a technical challenge due to chemical hallucinations look entirely plausible on the surface. Thirdly, there is a technical side effect observed in the research workflow regarding documentation. While the core scientific content differs, the heavy reliance on LLMs for drafting and structuring papers has led to an over-standardization of language, stripping away distinct nuances in how scientific research is communicated. Another connection to the Job Crafting Theory is Relational Crafting. Employees may interact differently with others as they can view that the journal papers done by coworkers follow the same AI structure, thus the use of AI in collaboration is understood, and communication between different teams.

In conclusion, this research provides insight into how generative AI affects workflow and collaboration in AstraZeneca Gothenburg, Sweden as a prime example. While GenAI offers proven productivity, its successful integration is context-based, such as task types, collaborations, and business context.

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